Coursework 2: Artificial Intelligence

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Question 1: Strategic Game Analysis

Part A: Utility Calculations

Expected utility in this context can be described as the net utility, which is equal to ***utility - cost.***

We use X, Y, and Z to refer to companies in each scenario because companies A, B, and C are interchangeable in this problem. Each scenario type can occur with different companies in different roles (e.g., A launches alone while B and C abstain, or B launches alone while A and C abstain), but the utilities remain the same due to symmetry. To avoid repetitive calculations, we present each unique scenario structure once using X, Y, Z notation.

Scenario 1: X launches alone while Y, Z abstain

$X_{expected\ utility}$ = *(utility of solo launch when others abstain) - (cost of solo launch)*
= 80 - 20
= 60

$Y_{expected\ utility} = Z_{expected\ utility}$ = (*utility of abstaining) - (cost of abstaining)*
= 0 - 0
= 0

Scenario 2: X, Y are part of Two-company Coalition, Z abstains

$X_{expected\ utility} = Y_{expected\ utility}$
= (utility of two-company coalition) – (coalition cost)
= (combined utility / $num_{companies}$) – (coalition cost)
= (120 / 2) – (10)
= 60 - 10
= 50

$Z_{expected\ utility}$ = (*utility of abstaining) - (cost of abstaining)*
= 0 - 0
= 0

Scenario 3: X, Y are part of Two-company Coalition, Z launches individually

$X_{expected\ utility} = Y_{expected\ utility}$
= (utility of two-company coalition) – (coalition cost)
= (combined utility / $num_{companies}$) – (coalition cost)
= (120 / 2) – (10)
= 60 - 10
= 50

$Z_{expected\ utility}$ = (*utility of solo launch when others also launch) - (cost of solo launch)*

= 50 - 20
= 30

Scenario 4: X,Y, Z are part of a Three-company Coalition

$X_{expected\ utility} = Y_{expected\ utility} = Z_{expected\ utility}$
= (utility of three-company coalition) – (coalition cost)
= (combined utility / $num_{companies}$) – (coalition cost)
= (150 / 3) – (10)
= 50 – 10
= 40

Scenario 5: X and Y launch individually, Z abstains

$X_{expected\ utility} = Y_{expected\ utility}$
= (utility of solo launch when others also launch) - (cost of solo launch)
= 50 - 20
= 30
$Z_{expected\ utility}$ = (*utility of abstaining) - (cost of abstaining)*
= 0 - 0
= 0

Scenario 6: X,Y, Z each launch individually

$X_{expected\ utility} = Y_{expected\ utility} = Z_{expected\ utility}$
= (utility of solo launch when others also launch) – (cost of solo launch)
= 50 – 20
= 30

Scenario 7: X,Y, Z all abstain

$X_{expected\ utility} = Y_{expected\ utility} = Z_{expected\ utility}$
= (*utility of abstaining) - (cost of abstaining)*
= 0 - 0
= 0

Part B: Nash Equilibrium

Yes, scenario 4 where X, Y, and Z form a three-company coalition is a **Nash equilibrium**. No player can improve their payoff by deviating. Each company has an expected utility of 40. Below I go through each scenario and explain why it is or is not a Nash Equilibrium:

Scenario 1: *X launches alone, Y and Z abstain*

Y or Z can deviate to launching alone (expected utility = 30) or forming a coalition with each other (expected utility = 50) which will be more profitable than 0.

Scenario 2: *X,Y coalition, Z abstains*
Z can deviate to launching alone (expected utility = 30) which will be more profitable than 0.

Scenario 3: *X,Y coalition, Z launches alone*
Z can deviate to joining a three-company coalition (expected utility = 40), which will be more profitable than 30 (existing utility).
*** T*his analysis assumes that coalition formation is an individual action available to each firm (i.e., firms may unilaterally choose to join a coalition).*

Scenario 4: *X,Y,Z three-company coalition -> **NASH EQUILIBRIUM***

- Any player deviating to a solo launch (expected utility = 30) will make less than 40 (existing utility).
- Any player deviating to abstaining (expected utility = 0) will make less than 40 (existing utility).

Therefore, no player can improve their payoff by deviating, which means this is a Nash equilibrium.

Scenario 5: *X,Y launch individually, Z abstains*
Z can deviate to launching alone (expected utility = 30) which will be more profitable than 0.
X or Y can deviate to forming a coalition (expected utility = 50) which is more profitable than 30 (existing utility)

Scenario 6: *X,Y,Z all launch individually*
Any player can deviate to forming a coalition, which is more profitable than 30 (existing utility)

Scenario 7: *X,Y,Z all abstain*
Any player can deviate to forming a coalition or launching alone (30), which is more profitable than 0 (existing utility)

Part C: Dominant Strategies

A strategy is dominant if it gives a player a higher payoff than any other strategy, regardless of what the other players do.

There is no dominant strategy because each player's best response depends on what the others do:

- Launching alone only has the highest payoff (utility = 60) if the other companies choose to abstain, which cannot be guaranteed.
- Abstaining has the lowest payoff (utility = 0), so it cannot be considered dominant.
- Forming a coalition if not possible if the other players choose to abstain or launch on their own.

Since the “best” strategy depends on what other players do, we can deduce that no dominant strategy exists.

Part D: Shapley Values

If all three companies form a grand coalition, **the Shapley value for each firm is 40.** The Shapley value is computed using net coalition values (after costs).

Let v(S) = the net utility a coalition S can generate:

- v({}) = 0 (empty coalition)
- v({A}) = v({B}) = v({C}) = (solo launch expected utility when no else has launched) - (solo launch cost) = 80 - 20 = 60
- v({A,B}) = v({A,C}) = v({B,C}) = (two-company coalition utility) - (total cost) = 120 - (10*2) = 100
- v({A,B,C}) = (three-company coalition utility) - (total cost) = 150 - (10*3) = 120 (grand coalition)

Calculating Marginal Contributions

We will calculate each company’s contribution depending on the order of joining. There are 3! = 6 possible orderings of companies joining:

<u>Order 1: A → B → C</u>
A joins empty: contribution = v({A}) - v({}) = 60 - 0 = 60
B joins A: contribution = v({A,B}) - v({A}) = 100 - 60 = 40
C joins A,B: contribution = v({A,B,C}) - v({A,B}) = 120 - 100 = 20

<u>Order 2: A → C → B</u>
A joins empty: contribution = v({A}) - v({}) = 60 - 0 = 60
C joins A: contribution = v({A,C}) - v({A}) = 100 - 60 = 40
B joins A,C: contribution = v({A,B,C}) - v({A,C}) = 120 - 100 = 20

<u>Order 3: B → A → C</u>
B joins empty: contribution = v({B}) - v({}) = 60 - 0 = 60
A joins B: contribution = v({A,B}) - v({B}) = 100 - 60 = 40
C joins A,B: contribution = v({A,B,C}) - v({A,B}) = 120 - 100 = 20

<u>Order 4: B → C → A</u>
B joins empty: contribution = v({B}) - v({}) = 60 - 0 = 60
C joins B: contribution = v({B,C}) - v({B}) = 100 - 60 = 40
A joins B,C: contribution = v({A,B,C}) - v({B,C}) = 120 - 100 = 20

<u>Order 5: C → A → B</u>
C joins empty: contribution = v({C}) - v({}) = 60 - 0 = 60
A joins C: contribution = v({A,C}) - v({C}) = 100 - 60 = 40
B joins A,C: contribution = v({A,B,C}) - v({A,C}) = 120 - 100 = 20

<u>Order 6: C → B → A</u>
C joins empty: contribution = v({C}) - v({}) = 60 - 0 = 60
B joins C: contribution = v({B,C}) - v({C}) = 100 - 60 = 40
A joins B,C: contribution = v({A,B,C}) - v({B,C}) = 120 - 100 = 20

Calculating Shapley Values

Shapley value = Average marginal contribution across all orderings

Company A:
Shapley value = (60 + 60 + 40 + 20 + 40 + 20) / 6 = 300 / 6 = 40

Company B:
Shapley value = (40 + 20 + 60 + 60 + 20 + 40) / 6 = 300 / 6 = 40

Company C:
Shapley value = (20 + 40 + 20 + 40 + 60 + 60) / 6 = 300 / 6 = 40

Thus, the Shapley value for each company is **40.**

Question 2: Decision Tree Design with ID3

Part A: Scenario and Features

Scenario: I need to decide whether to head to the pub or stay home. This classification task predicts my social behavior based on academic progress, financial status, location, and physical readiness. The **target class label** is "Go Out" (Binary: Yes/No).

Input Variables:

- Assignments Completed (AC): A binary feature indicating if all pending assignments for the week have been completed [Yes, No]
- Account Balance (AB): An ordinal ternary feature representing the amount of disposable money I have in my account [Low (<£30), Mid (£30–£100), High (>£100)]
- City (C): A ternary feature representing my physical location, which influences the social circle I'll be seeing [London, Birmingham, Oxford]
- Eaten Food (F): A binary feature indicating if I have eaten dinner or not, which can influence energy levels and the ability to hold alcohol [Yes, No]

Training Examples:

ID	Assignments Completed (AC)	Account Balance (AB)	City (C)	Eaten Food (F)	Go Out?
1	Yes	High	London	Yes	Yes
2	Yes	Mid	Birmingham	Yes	No
3	Yes	Low	Oxford	No	No
4	No	High	London	No	No
5	Yes	Mid	London	No	Yes
6	No	Low	Birmingham	Yes	No
7	Yes	High	Oxford	Yes	Yes
8	Yes	Mid	Birmingham	No	Yes
9	No	Mid	London	Yes	Yes
10	Yes	High	Birmingham	Yes	Yes
11	Yes	Low	London	No	No
12	No	High	Oxford	Yes	No
13	Yes	Mid	Oxford	Yes	Yes
14	Yes	High	London	No	Yes
15	No	Mid	Birmingham	No	No
16	Yes	Low	Birmingham	Yes	Yes
17	Yes	High	Oxford	No	Yes

18	No	High	London	Yes	Yes
19	Yes	Mid	London	Yes	Yes
20	Yes	High	Birmingham	No	No
21	No	High	Birmingham	No	No
22	Yes	Mid	Oxford	No	No
23	Yes	Low	Birmingham	No	No

Part B: Entropy and Root Information Gain

Class Entropy H(S)

Target Class Counts:

- $\left|S_{yes}\right| = 12$
- $\left|S_{no}\right| = 11$
- $|S| = S_{yes} + S_{no} = 23$

Target Class Probabilities

- $P(Yes) = \frac{\left|S_{yes}\right|}{|S|} = \frac{12}{23} \approx 0.5217$
- $P(No) = \frac{\left|S_{no}\right|}{|S|} = \frac{11}{23} \approx 0.4783$

Conditional Entropy Calculation

$$H(S) = -\sum_{x \epsilon S} P(x)\, log_2 P(x)$$

$$= -[P(Yes)\, log_2 P(Yes) + P(No)\, log_2 P(No)]$$

$$= -[\frac{12}{23}\, log_2 \frac{12}{23} + \frac{11}{23}\, log_2 \frac{11}{23}]$$

$$\approx 0.9986$$

Information Gain IG

$IG(S, Q) = H(S) - \sum_{S_i \epsilon Split(S, Q)} \frac{\left|S_i\right|}{|S|} H(S_i)$, where Q is a feature you split on

<u>Assignments Completed (AC)</u>

Class Counts:

- Group: $AC = Yes$ (Total: 16 rows)
  - $Go\ Out = Yes$: 10
  - $Go\ Out = No$: 6
- Group: $AC = No$ (Total: 7 rows)
  - $Go\ Out = Yes$: 2
  - $Go\ Out = No$: 5

Conditional Class Probabilities:

- $P(GoOut = Yes\,|AC = Yes) = \frac{|S_{GoOut=Yes\ and\ AC=Yes}|}{|S_{AC=Yes}|} = \frac{10}{16} = 0.6250$
- $P(GoOut = No\,|AC = Yes) = \frac{|S_{GoOut=No\ and\ AC=Yes}|}{|S_{AC=Yes}|} = \frac{6}{16} = 0.3750$
- $P(GoOut = Yes\,|AC = No) = \frac{|S_{GoOut=Yes\ and\ AC=No}|}{|S_{AC=No}|} = \frac{2}{7} \approx 0.2857$
- $P(GoOut = No\,|AC = No) = \frac{|S_{GoOut=No\ and\ AC=No}|}{|S_{AC=No}|} = \frac{5}{7} \approx 0.7143$

Conditional Class Entropy

$$H(S_{AC=Yes}) = -P(GoOut = Yes\,|AC = Yes)\,log_2 P(GoOut = Yes\,|AC = Yes)$$
$$-P(GoOut = No\,|AC = Yes)\,log_2 P(GoOut = No\,|AC = Yes)$$

$$= -\frac{10}{16} log_2 \frac{10}{16} - \frac{6}{16} log_2 \frac{6}{16}$$

$$\approx 0.9544$$

$$H(S_{AC=No}) = -P(GoOut = Yes\,|AC = No)\,log_2 P(GoOut = Yes\,|AC = No)$$
$$-P(GoOut = No\,|AC = No)\,log_2 P(GoOut = No\,|AC = No)]$$

$$= -\frac{2}{7} log_2 \frac{2}{7} - \frac{5}{7} log_2 \frac{5}{7}$$

$$\approx 0.8631$$

Information Gain

$$IG(S, AC) = H(S) - \frac{|S_{AC=Yes}|}{|S|} H(S_{AC=Yes}) - \frac{|S_{AC=No}|}{|S|} H(S_{AC=No})$$

$$\approx 0.9986 - \frac{16}{23}(0.9544) - \frac{7}{23}(0.8631)$$

$$\approx 0.072$$

Account Balance (AB)

Class Counts:

- Group: $AB = Low$ (Total: 5 rows)
  - $Go\ Out = Yes$: 1
  - $Go\ Out = No$: 4
- Group: $AB = Mid$ (Total: 8 rows)
  - $Go\ Out = Yes$: 5
  - $Go\ Out = No$: 3
- Group: $AB = High$ (Total: 10 rows)
  - $Go\ Out = Yes$: 6
  - $Go\ Out = No$: 4

Conditional Class Probabilities:

- $P(GoOut = Yes\,|AB = Low) = \frac{|S_{GoOut=Yes\ and\ AB=Low}|}{|S_{AB=Low}|} = \frac{1}{5} = 0.2000$
- $P(GoOut = No\,|AB = Low) = \frac{|S_{GoOut=No\ and\ AB=Low}|}{|S_{AB=Low}|} = \frac{4}{5} = 0.8000$
- $P(GoOut = Yes\,|AB = Mid) = \frac{|S_{GoOut=Yes\ and\ AB=Mid}|}{|S_{AB=Mid}|} = \frac{5}{8} = 0.6250$
- $P(GoOut = No\,|AB = Mid) = \frac{|S_{GoOut=No\ and\ AB=Mid}|}{|S_{AB=Mid}|} = \frac{3}{8} = 0.3750$
- $P(GoOut = Yes\,|AB = High) = \frac{|S_{GoOut=Yes\ and\ AB=High}|}{|S_{AB=High}|} = \frac{6}{10} \approx 0.6000$
- $P(GoOut = No\,|AB = High) = \frac{|S_{GoOut=No\ and\ AB=High}|}{|S_{AB=High}|} = \frac{4}{10} \approx 0.4000$

Conditional Class Entropy

$H(S_{AB=Low}) = \ - P(GoOut = Yes\,|AB = Low)\, log_2 P(GoOut = Yes\,|AB = Low)$

$- P(GoOut = No\,|AB = Low)\, log_2 P(GoOut = No\,|AB = Low)$

= $- 0.2\, log_2 0.2 \ - 0.8\, log_2 0.8$

≈ 0.7219

$H(S_{AB=Mid}) = \ - P(GoOut = Yes\,|AB = Mid)\, log_2 P(GoOut = Yes\,|AB = Mid)$

$- P(GoOut = No\,|AB = Mid)\, log_2 P(GoOut = No\,|AB = Mid)]$

$- 0.625\, log_2 0.625 \ - 0.375\, log_2 0.375$

$\approx$ 0.9544

$H(S_{AB=High}) = \; - P(GoOut = Yes\,|AB = High)\, log_2 P(GoOut = Yes\,|AB = High)$

$- P(GoOut = No\,|AB = High)\, log_2 P(GoOut = No\,|AB = High)]$

$= - 0.6\, log_2 0.6 \; - 0.4\, log_2 0.4$

$\approx$ 0.9710

Information Gain

$IG(S,\, AB) = H(S) - \frac{|S_{AB=Low}|}{|S|} H(S_{AB=Low}) - \frac{|S_{AB=Mid}|}{|S|} H(S_{AB=Mid}) - \frac{|S_{AB=High}|}{|S|} H(S_{AB=High})$

$\approx 0.9986 - \frac{5}{23}(0.7219) - \frac{8}{23}(0.9544) - \frac{10}{23}(0.9710)$

≈ 0.0875

City (C)

Class Counts:

- Group: $C = London$ (Total: 8 rows)
  - $Go\ Out = Yes$: 6
  - $Go\ Out = No$: 2
- Group: $C = Birmingham$ (Total: 9 rows)
  - $Go\ Out = Yes$: 3
  - $Go\ Out = No$: 6
- Group: $C = Oxford$ (Total: 6 rows)
  - $Go\ Out = Yes$: 3
  - $Go\ Out = No$: 3

Conditional Class Probabilities:

- $P(GoOut = Yes\,|C = London) = \frac{|S_{GoOut=Yes\ and\ C=London}|}{|S_{C=London}|} = \frac{6}{8} = 0.7500$
- $P(GoOut = No\,|C = London) = \frac{|S_{GoOut=No\ and\ C=London}|}{|S_{C=London}|} = \frac{2}{8} = 0.2500$
- $P(GoOut = Yes\,|C = Birmingham) = \frac{|S_{GoOut=Yes\ and\ C=Birmingham}|}{|S_{C=Birmingham}|} = \frac{3}{9} \approx 0.3333$
- $P(GoOut = No\,|C = Birmingham) = \frac{|S_{GoOut=No\ and\ C=Birmingham}|}{|S_{C=Birmingham}|} = \frac{6}{9} = 0.6667$
- $P(GoOut = Yes\,|C = Oxford) = \frac{|S_{GoOut=Yes\ and\ C=Oxford}|}{|S_{C=Oxford}|} = \frac{3}{6} \approx 0.5000$

- $P(GoOut = No \mid C = Oxford) = \frac{|S_{GoOut=No\ and\ C=Oxford}|}{|S_{C=Oxford}|} = \frac{3}{6} \approx 0.5000$

Conditional Class Entropy

$H(S_{C=London}) = \ - P(GoOut = Yes \mid C = London)\, log_2 P(GoOut = Yes \mid C = London)$

$- P(GoOut = No \mid C = London)\, log_2 P(GoOut = No \mid C = London)$

$= - 0.75\, log_2 0.75 \ - 0.25\, log_2 0.25$

≈ 0.8113

$H(S_{C=Birmingham}) =- P(GoOut = Yes \mid C = Birmingham)\, log_2 P(GoOut = Yes \mid C = Birmingham)$

$- P(GoOut = No \mid C = Birmingham)\, log_2 P(GoOut = No \mid C = Birmingham)]$

$- \frac{3}{9}\, log_2 \frac{3}{9} \ - \frac{6}{9}\, log_2 \frac{6}{9}$

≈ 0.9183

$H(S_{C=Oxford}) = \ - P(GoOut = Yes \mid C = Oxford)\, log_2 P(GoOut = Yes \mid C = Oxford)$

$- P(GoOut = No \mid C = Oxford)\, log_2 P(GoOut = No \mid C = Oxford]$

$= - 0.5\, log_2 0.5 \ - 0.5\, log_2 0.5$

= 1

Information Gain

$IG(S,\ C) = H(S) - \frac{|S_{C=London}|}{|S|} H(S_{C=London}) - \frac{|S_{C=Birmingham}|}{|S|} H(S_{C=Birmingham}) - \frac{|S_{C=Oxford}|}{|S|} H(S_{C=Oxford})$

$\approx 0.9986 - \frac{8}{23}(0.8113) - \frac{9}{23}(0.9183) - \frac{6}{23}(1)$

≈ 0.0962

<u>Eaten Food (F)</u>

Class Counts:

- Group: $F = Yes$ (Total: 11 rows)
  - $Go\ Out = Yes$: 8
  - $Go\ Out = No$: 3
- Group: $F = No$ (Total: 12 rows)
  - $Go\ Out = Yes$: 4
  - $Go\ Out = No$: 8

Conditional Class Probabilities:

- $P(GoOut = Yes \,|F = Yes) = \frac{|S_{GoOut=Yes\ and\ F=Yes}|}{|S_{F=Yes}|} = \frac{8}{11} \approx 0.7273$
- $P(GoOut = No \,|F = Yes) = \frac{|S_{GoOut=No\ and\ F=Yes}|}{|S_{F=Yes}|} = \frac{3}{11} \approx 0.2727$
- $P(GoOut = Yes \,|F = No) = \frac{|S_{GoOut=Yes\ and\ F=No}|}{|S_{F=No}|} = \frac{4}{12} \approx 0.3333$
- $P(GoOut = No \,|F = No) = \frac{|S_{GoOut=No\ and\ F=No}|}{|S_{F=No}|} = \frac{8}{12} \approx 0.6667$

Conditional Class Entropy

$H(S_{F=Yes}) = \ - P(GoOut = Yes \,|F = Yes)\, log_2 P(GoOut = Yes \,|F = Yes)$

$- P(GoOut = No \,|F = Yes)\, log_2 P(GoOut = No \,|F = Yes)$

$= -\frac{8}{11} log_2 \frac{8}{11} \ - \frac{3}{11} log_2 \frac{3}{11}$

≈ 0.8454

$H(S_{F=No}) = \ - P(GoOut = Yes \,|F = No)\, log_2 P(GoOut = Yes \,|F = No)$

$- P(GoOut = No \,|F = No)\, log_2 P(GoOut = No \,|F = No)]$

$= -\frac{4}{12} log_2 \frac{4}{12} \ - \frac{8}{12} log_2 \frac{8}{12}$

≈ 0.9183

Information Gain

$IG(S, F) = H(S) - \frac{|S_{F=Yes}|}{|S|} H(S_{F=Yes}) - \frac{|S_{F=No}|}{|S|} H(S_{F=No})$

$\approx 0.9986 - \frac{11}{23}(0.8454) - \frac{12}{23}(0.9183)$

≈ 0.1152

Feature with the Highest Information Gain

The feature with the highest information gain is F, which represents “Eaten Food”, with an Information Gain of 0.1152. Therefore, “Eaten Food” will be chosen as the root node of the decision tree.

$IG_F = 0.1152 > IG_C = 0.0962 > IG_{AB} = 0.0875 > IG_{AC} = 0.072$

Part C: Recursive Splitting

Level 1: Split on F

Branch 1a: F=Yes (11 instances)

Indices in this branch: [1, 2, 6, 7, 9, 10, 12, 13, 16, 18, 19]

Class Counts

- Go Out = Yes: 8 (rows 1, 7, 9, 10, 13, 16, 18, 19)
- Go Out = No: 3 (rows 2, 6, 12)

Entropy

$H(S_{F=Yes}) =- [P(Go\ Out\ =\ Yes)\ log_2 P(Go\ Out\ =\ Yes)\ +\ P(Go\ Out\ =\ No)\ log_2 P(Go\ Out\ =\ No)]$

$=- [\frac{8}{11} log_2 \frac{8}{11}\ + \frac{3}{11}\ log_2 \frac{3}{11}]$

$\approx\ 0.8454$

Feature AC

- Counts:
  - Group: $AC\ =\ Yes$ (7 rows: [1, 2, 7, 10, 13, 16, 19])
    - Go Out = Yes: 6 (rows 1, 7, 10, 13, 16, 19)
    - Go Out = No: 1 (rows 2)
  - Group: $AC\ =\ No$ (4 rows: [6, 9, 12, 18])
    - Go Out = Yes: 2 (rows 9, 18)
    - Go Out = No: 2 (rows 6, 12)
- $H(S_{F=Yes,\ AC=Yes}) =\ -\frac{6}{7} log_2 \frac{6}{7}\ -\frac{1}{7} log_2 \frac{1}{7} \approx\ 0.5917$
- $H(S_{F=Yes,\ AC=No}) =\ -\frac{2}{4} log_2 \frac{2}{4}\ -\frac{2}{4} log_2 \frac{2}{4} = 1$
- $IG(S_{F=Yes},\ AC) =\ 0.8454\ -\frac{7}{11}(0.5917)\ -\frac{4}{11}(1)\ =\ 0.1052$

Feature AB

- Counts:
  - Group: $AB\ =\ High$ (5 rows: [1, 7, 12, 10, 18])
    - Go Out = Yes: 4 (rows 1, 7, 10, 18)
    - Go Out = No: 1 (row 12)
  - Group: $AB\ =\ Mid$ (4 rows: [2, 9, 13, 19])
    - Go Out = Yes: 3 (rows 9, 13, 19)
    - Go Out = No: 1 (row 2)
  - Group: $AB\ =\ Low$ (2 rows: [6, 16])
    - Go Out = Yes: 1 (row 16)
    - Go Out = No: 1 (row 6)
- $H(S_{F=Yes,\ AB=High}) =\ -\frac{4}{5} log_2 \frac{4}{5}\ -\frac{1}{5} log_2 \frac{1}{5} \approx\ 0.7219$

- $H(S_{F=Yes,\, AB=Mid}) = -\frac{3}{4}log_2\frac{3}{4} - \frac{1}{4}log_2\frac{1}{4} \approx 0.8113$
- $H(S_{F=Yes,\, AB=Low}) = -\frac{1}{2}log_2\frac{1}{2} - \frac{1}{2}log_2\frac{1}{2} = 1$
- $IG(S_{F=Yes},\, AB) = 0.8454 - \frac{5}{11}(0.7219) - \frac{4}{11}(0.8113) - \frac{2}{11}(1) = 0.0404$

Feature C

- Counts:
  - Group: $C = London$ (4 rows: [1, 9, 18, 19])
    - Go Out = Yes: 4 (rows 1, 9, 18, 19)
    - Go Out = No: 0
  - Group: $C = Birmingham$ (4 rows: [2, 6, 10, 16])
    - Go Out = Yes: 2 (rows 10, 16)
    - Go Out = No: 2 (rows 2, 6)
  - Group: $C = Oxford$ (3 rows: [7, 12, 13])
    - Go Out = Yes: 2 (rows 7, 13)
    - Go Out = No: 1 (row 12)
- $H(S_{F=Yes,\, C=London}) = -\frac{4}{4}log_2\frac{4}{4} - \frac{0}{4}log_2\frac{0}{4} = 0$
- $H(S_{F=Yes,\, C=Birmingham}) = -\frac{2}{4}log_2\frac{2}{4} - \frac{2}{4}log_2\frac{2}{4} = 1$
- $H(S_{F=Yes,\, C=Oxford}) = -\frac{2}{3}log_2\frac{2}{3} - \frac{1}{3}log_2\frac{1}{3} \approx 0.9183$
- $IG(S_{F=Yes},\, C) = 0.8454 - \frac{4}{11}(0) - \frac{4}{11}(1) - \frac{3}{11}(0.9183) = 0.2313$

Splitting on Feature C next, as IG = 0.2313 is highest.

Level 2: Split on C (From F=Yes)

Branch 1a.2a: C=London (4 instances)

Indices in this branch: [1, 9, 18, 19]

Class Counts

- Go Out = Yes: 4 (rows 1, 9, 18, 19)
- Go Out = No: 0

LEAF NODE!

Branch 1a.2b: C=Birmingham (4 instances)

Indices in this branch: [2, 6, 10, 16]

Class Counts

- Go Out = Yes: 2 (rows 10, 16)
- Go Out = No: 2 (rows 2, 6)

Entropy

$H(S_{F=Yes,\, C=Birmingham}) = -[\frac{2}{4}log_2\frac{2}{4} + \frac{2}{4}\, log_2\frac{2}{4}] = 1$

Feature AC

- Counts:

- Group: $AC = Yes$ (3 rows: [2, 10, 16])
  - Go Out = Yes: 2 (rows 10, 16)
  - Go Out = No: 1 (row 2)
- Group: $AC = No$ (1 row: [6])
  - Go Out = Yes: 0
  - Go Out = No: 1 (row 6)

- $H(S_{F=Yes,\, C=Birmingham, AC=Yes}) = -\frac{2}{3}log_2\frac{2}{3} - \frac{1}{3}log_2\frac{1}{3} \approx 0.9183$
- $H(S_{F=Yes, C=Birmingham,\, AC=No}) = -\frac{0}{1}log_2\frac{0}{1} - \frac{1}{1}log_2\frac{1}{1} = 0$
- $IG(S_{F=Yes,\, C=Birmingham}, AC) = 1 - \frac{3}{4}(0.9183) - \frac{1}{4}(0) \approx 0.3113$

Feature AB

- Counts:
  - Group: $AB = High$ (1 row: [10])
    - Go Out = Yes: 1 (row 10)
    - Go Out = No: 0
  - Group: $AB = Mid$ (1 row: [2])
    - Go Out = Yes: 0
    - Go Out = No: 1 (row 2)
  - Group: $AB = Low$ (2 rows: [6, 16])
    - Go Out = Yes: 1 (row 16)
    - Go Out = No: 1 (row 6)

- $H(S_{F=Yes,\, C=Birmingham,\, AB=High}) = -\frac{1}{1}log_2\frac{1}{1} - \frac{0}{1}log_2\frac{0}{1} = 0$
- $H(S_{F=Yes,\, C=Birmingham,\, AB=Mid}) = -\frac{0}{1}log_2\frac{0}{1} - \frac{1}{1}log_2\frac{1}{1} = 0$
- $H(S_{F=Yes,\, C=Birmingham,\, AB=Low}) = -\frac{1}{2}log_2\frac{1}{2} - \frac{1}{2}log_2\frac{1}{2} = 1$
- $IG(S_{F=Yes,\, C=Birmingham}, AB) = 1 - \frac{1}{4}(0) - \frac{1}{4}(0) - \frac{2}{4}(1) = 0.5$

Splitting on Feature AB next, as IG = 0.5 is highest.

Level 3: Split on AB (From F=Yes, C=Birmingham)

Branch 1a.2b.3a: AB=High (1 instance)

Indices in this branch: [10]

Class Counts

- Go Out = Yes: 1 (row 10)
- Go Out = No: 0

LEAF NODE!

Branch 1a.2b.3b: AB=Mid (1 instance)

Indices in this branch: [2]

Class Counts

- Go Out = Yes: 0
- Go Out = No: 1 (row 2)

LEAF NODE!

Branch 1a.2b.3c: AB=Low (2 instances)

Indices in this branch: [6, 16]

Class Counts

- Go Out = Yes: 1 (row 16)
- Go Out = No: 1 (row 6)

Entropy

$H(S_{F=Yes,\, C=Birmingham, AB=Low}) = \; -[\frac{1}{2} log_2 \frac{1}{2} \; + \frac{1}{1} \; log_2 \frac{1}{2}] = 1$

Feature AC

- Counts:
  - Group: $AC = Yes$ (1 row: [16])
    - Go Out = Yes: 1 (row 16)
    - Go Out = No: 0
  - Group: $AC = No$ (1 row: [6])
    - Go Out = Yes: 0
    - Go Out = No: 1 (row 6)
- $H(S_{F=Yes,\, C=Birmingham, AB=Low,\, AC=Yes}) = \; -\frac{1}{1} log_2 \frac{1}{1} \; - \frac{0}{1} log_2 \frac{0}{1} = 0$
- $H(S_{F=Yes, C=Birmingham,\, AB=Low,\, AC=No}) = \; -\frac{0}{1} log_2 \frac{0}{1} \; - \frac{1}{1} log_2 \frac{1}{1} = 0$
- $IG(S_{F=Yes,\, C=Birmingham,\, AB=Low}, AC) = 1 \; - \frac{1}{2}(0) - \frac{1}{2}(0) = 1$

Splitting on Feature AC next, as it is our only option

Level 4: Split on AC (From F=Yes, C=Birmingham, AB=Low)

Branch 1a.2b.3c.4a: AC=Yes (1 instance)

Indices in this branch: [16]

Class Counts

- Go Out = Yes: 1 (row 16)
- Go Out = No: 0

LEAF NODE!

Branch 1a.2b.3a.4b: AC=No (1 instance)

Indices in this branch: [6]

Class Counts

- Go Out = Yes: 0
- Go Out = No: 1 (row 6)

LEAF NODE!

Branch 1a.2c: C=Oxford (3 instances)

Indices in this branch: [7, 12, 13]

Class Counts

- Go Out = Yes: 2 (rows 7, 13)
- Go Out = No: 1 (row 12)

Entropy

$$H(S_{F=Yes,\,C=Oxford}) = -[\tfrac{2}{3}log_2\tfrac{2}{3} + \tfrac{1}{3}\,log_2\tfrac{1}{3}] = 0.9183$$

Feature AC

- Counts:
  - Group: $AC = Yes$ (2 rows: [7, 13])
    - Go Out = Yes: 2 (rows 7, 13)
    - Go Out = No: 0
  - Group: $AC = No$ (1 row: [12])
    - Go Out = Yes: 0
    - Go Out = No: 1 (row 12)
- $H(S_{F=Yes,\,C=Oxford,AC=Yes}) = -\frac{2}{2}log_2\frac{2}{2} - \frac{0}{2}log_2\frac{0}{2} = 0$
- $H(S_{F=Yes,C=Oxford,\,AC=No}) = -\frac{1}{1}log_2\frac{1}{1} - \frac{0}{1}log_2\frac{0}{1} = 0$
- $IG(S_{F=Yes,\,C=Oxford},\,AC) = 0.9183 - \frac{2}{3}(0) - \frac{1}{3}(0) \approx 0.9183$

Feature AB

- Counts:
  - Group: $AB = High$ (2 rows: [7, 12])
    - Go Out = Yes: 1 (row 7)
    - Go Out = No: 1 (row 12)
  - Group: $AB = Mid$ (1 row: [13])
    - Go Out = Yes: 1 (row 13)
    - Go Out = No: 0
  - Group: $AB = Low$ (0 rows: [])
    - Go Out = Yes: 0
    - Go Out = No: 0
- $H(S_{F=Yes,\,C=Oxford,\,AB=High}) = -\frac{1}{2}log_2\frac{1}{2} - \frac{1}{2}log_2\frac{1}{2} = 1$
- $H(S_{F=Yes,\,C=Oxford,\,AB=Mid}) = -\frac{1}{1}log_2\frac{1}{1} - \frac{0}{1}log_2\frac{0}{1} = 0$
- $H(S_{F=Yes,\,C=Oxford,\,AB=Low}) = 0$
- $IG(S_{F=Yes,\,C=Oxford},\,AB) = 0.9183 - \frac{2}{3}(1) - \frac{1}{3}(0) - 0 = 0.2516$

Splitting on Feature AC next, as IG = 0.9183 is highest.

Level 3: Split on AC (From F=Yes, C=Oxford)

Branch 1a.2c.3a: AC=Yes (2 instances)

Indices in this branch: [7, 13]

Class Counts

- Go Out = Yes: 2 (rows 7, 13)
- Go Out = No: 0

LEAF NODE!

Branch 1a.2c.3b: AC=No (1 instance)

Indices in this branch: [12]

Class Counts

- Go Out = Yes: 0
- Go Out = No: 1 (row 12)

LEAF NODE!

Branch 1b: F=No (12 instances)

Indices in this branch: [3, 4, 5, 8, 11, 14, 15, 17, 20, 21, 22, 23]

Class Counts

- Go Out = Yes: 4 (rows 5, 8, 14, 17)
- Go Out = No: 8 (rows 3, 4, 11, 15, 20, 21, 22, 23)

Entropy

$H(S_{F=No}) =- [P(Go\ Out\ = Yes)\ log_2 P(Go\ Out\ = Yes)\ + P(Go\ Out\ = No)\ log_2 P(Go\ Out\ = No)]$

$=- [\frac{4}{12} log_2 \frac{4}{12}\ + \frac{8}{12}\ log_2 \frac{8}{12}] \approx\ 0.9183$

Feature AC

- Counts:
  - Group: $AC\ =\ Yes$ (9 rows: [3, 5, 8, 11, 14, 17, 20, 22, 23])
    - Go Out = Yes: 4 (rows 5, 8, 14, 17)
    - Go Out = No: 5 (rows 3, 11, 20, 22, 23)
  - Group: $AC\ =\ No$ (3 rows: [4, 15, 21])
    - Go Out = Yes: 0
    - Go Out = No: 3 (rows 4, 15, 21)
- $H(S_{F=No,\ AC=Yes}) =\ \ -\frac{4}{9} log_2 \frac{4}{9}\ \ -\frac{5}{9} log_2 \frac{5}{9} \approx\ 0.9911$
- $H(S_{F=No,\ AC=No}) =\ \ -\frac{0}{3} log_2 \frac{0}{3}\ \ -\frac{3}{3} log_2 \frac{3}{3} =\ 0$
- $IG(S_{F=No},\ AC) =\ 0.9183\ \ -\frac{9}{12}(0.9911)\ -\frac{3}{12}(0) =\ 0.1750$

Feature AB

- Counts:
  - Group: $AB\ =\ High$ (5 rows: [4, 14, 17, 20, 21])
    - Go Out = Yes: 2 (rows 14, 17)
    - Go Out = No: 3 (rows 4, 20, 21)
  - Group: $AB\ =\ Mid$ (4 rows: [5, 8, 15, 22])

        - Go Out = Yes: 2 (rows 5, 8)
        - Go Out = No: 2 (rows 15, 22)
    - Group: $AB = Low$ (3 rows: [3, 11, 23])
        - Go Out = Yes: 0
        - Go Out = No: 3 (rows 3, 11, 23)
- $H(S_{F=No,\, AB=High}) = -\frac{2}{5}log_2\frac{2}{5} - \frac{3}{5}log_2\frac{3}{5} \approx 0.9710$
- $H(S_{F=No,\, AB=Mid}) = -\frac{2}{4}log_2\frac{2}{4} - \frac{2}{4}log_2\frac{2}{4} = 1$
- $H(S_{F=No,\, AB=Low}) = -\frac{0}{3}log_2\frac{0}{3} - \frac{3}{3}log_2\frac{3}{3} = 0$
- $IG(S_{F=No},\, AB) = 0.9183 - \frac{5}{12}(0.9710) - \frac{4}{12}(1) - \frac{3}{12}(0) = 0.1804$

Feature C

- Counts:
    - Group: $C = London$ (4 rows: [4, 5, 11, 14])
        - Go Out = Yes: 2 (rows 5, 14)
        - Go Out = No: 2 (rows 4, 11)
    - Group: $C = Birmingham$ (5 rows: [8, 15, 20, 21, 23])
        - Go Out = Yes: 1 (row 8)
        - Go Out = No: 4 (rows 15, 20, 21, 23)
    - Group: $C = Oxford$ (3 rows: [3, 17, 22])
        - Go Out = Yes: 1 (row 17)
        - Go Out = No: 2 (rows 3, 22)
- $H(S_{F=No,\, C=London}) = -\frac{2}{4}log_2\frac{2}{4} - \frac{2}{4}log_2\frac{2}{4} = 1$
- $H(S_{F=No,\, C=Birmingham}) = -\frac{1}{5}log_2\frac{1}{5} - \frac{4}{5}log_2\frac{4}{5} \approx 0.7219$
- $H(S_{F=No,\, C=Oxford}) = -\frac{1}{3}log_2\frac{1}{3} - \frac{2}{3}log_2\frac{2}{3} \approx 0.9183$
- $IG(S_{F=No},\, C) = 0.9183 - \frac{4}{12}(1) - \frac{5}{12}(0.7219) - \frac{3}{12}(0.9183) = 0.0546$

Splitting on Feature AB next, as IG = 0.1804 is highest.

Level 2: Split on AB (From F=No)

Branch 1b.2a: AB=High (5 instances)

Indices in this branch: [4, 14, 17, 20, 21]

Class Counts

- Go Out = Yes: 2 (rows 14, 17)
- Go Out = No: 3 (rows 4, 20, 21)

Entropy

$H(S_{F=No,\, AB=High}) = -[\frac{2}{5}log_2\frac{2}{5} + \frac{3}{5}\, log_2\frac{3}{5}] = 0.9710$

Feature AC

- Counts:
  - Group: $AC = Yes$ (3 rows: [14, 17, 20])
    - Go Out = Yes: 2 (rows 14, 17)
    - Go Out = No: 1 (row 20)
  - Group: $AC = No$ (2 rows: [4, 21])
    - Go Out = Yes: 0
    - Go Out = No: 2 (rows 4, 21)
- $H(S_{F=No,\, AB=High,\, AC=Yes}) = -\frac{2}{3}log_2\frac{2}{3} - \frac{1}{3}log_2\frac{1}{3} = 0.9183$
- $H(S_{F=No,\, AB=High,\, AC=No}) = -\frac{0}{2}log_2\frac{0}{2} - \frac{2}{2}log_2\frac{2}{2} = 0$
- $IG(S_{F=No,\, AB=High}, AC) = 0.9710 - \frac{3}{5}(0.9183) - \frac{2}{5}(0) = 0.4200$

Feature C

- Counts:
  - Group: $C = London$ (2 rows: [4, 14])
    - Go Out = Yes: 1 (row 14)
    - Go Out = No: 1 (row 4)
  - Group: $C = Birmingham$ (2 rows: [20, 21])
    - Go Out = Yes: 0
    - Go Out = No: 2 (rows 20, 21)
  - Group: $C = Oxford$ (1 row: [17])
    - Go Out = Yes: 1 (row 17)
    - Go Out = No: 0
- $H(S_{F=No,\, AB=High,\, C=London}) = -\frac{1}{2}log_2\frac{1}{2} - \frac{1}{2}log_2\frac{1}{2} = 1$
- $H(S_{F=No,\, AB=High,\, C=Birmingham}) = -\frac{0}{2}log_2\frac{0}{2} - \frac{2}{2}log_2\frac{2}{2} = 0$
- $H(S_{F=No,\, AB=High,\, C=Oxford}) = -\frac{1}{1}log_2\frac{1}{1} - \frac{0}{1}log_2\frac{0}{1} = 0$
- $IG(S_{F=No,\, AB=High}, C) = 0.9710 - \frac{2}{5}(1) - \frac{2}{5}(0) - \frac{1}{5}(0) = 0.5710$

Splitting on Feature C next, as IG = 0.5710 is highest.

Level 3: Split on C (From F=No, AB=High)

Branch 1b.2a.3a: C=London (2 instances)

Indices in this branch: [4, 14]

Class Counts

- Go Out = Yes: 1 (row 14)
- Go Out = No: 1 (row 4)

Entropy

$H(S_{F=No,\, AB=High,\, City=London}) = -[\frac{1}{2}log_2\frac{1}{2} + \frac{1}{1}\, log_2\frac{1}{2}] = 1$

Feature AC

- Counts:

- Group: $AC = Yes$ (1 row: [14])
  - Go Out = Yes: 1 (row 14)
  - Go Out = No: 0
- Group: $AC = No$ (1 row: [4])
  - Go Out = Yes: 0
  - Go Out = No: 1 (row 4)

- $H(S_{F=No,\, AB=High,\, City=London,\, AC=Yes}) = -\frac{1}{1}log_2\frac{2}{3} - \frac{0}{1}log_2\frac{1}{3} = 0$
- $H(S_{F=No,\, AB=High,\, City=London,\, AC=No}) = -\frac{0}{1}log_2\frac{0}{1} - \frac{1}{1}log_2\frac{1}{1} = 0$
- $IG(S_{F=No,\, AB=High,\, City=London}, AC) = 1 - \frac{1}{2}(0) - \frac{1}{2}(0) = 1$

Splitting on Feature AC next, as it is our only option

Level 4: Split on AC (From F=No, AB=High, C=London)

Branch 1b.2a.3a.4a: AC=Yes (1 instance)

Indices in this branch: [14]

Class Counts

- Go Out = Yes: 1 (row 14)
- Go Out = No: 0

LEAF NODE!

Branch 1b.2a.3a.4b: AC=No (1 instance)

Indices in this branch: [4]

Class Counts

- Go Out = Yes: 0
- Go Out = No: 1 (row 4)

LEAF NODE!

Branch 1b.2a.3b: C=Birmingham (2 instances)

Indices in this branch: [20, 21]

Class Counts

- Go Out = Yes: 0
- Go Out = No: 2 (rows 20, 21)

LEAF NODE!

Branch 1b.2a.3b: C=Oxford (1 instance)

Indices in this branch: [17]

Class Counts

- Go Out = Yes: 1 (row 17)
- Go Out = No: 0

LEAF NODE!

Branch 1b.2b: AB=Mid (4 instances)

Indices in this branch: [5, 8, 15, 22]

Class Counts

- Go Out = Yes: 2 (rows 5, 8)
- Go Out = No: 2 (rows 15, 22)

Entropy

$H(S_{F=No,\,AB=Mid}) = -[\frac{2}{4}log_2\frac{2}{4} + \frac{2}{4}\,log_2\frac{2}{4}] = 1$

Feature AC

- Counts:
  - Group: $AC = Yes$ (3 rows: [5, 8, 22])
    - Go Out = Yes: 2 (rows 5, 8)
    - Go Out = No: 1 (row 22)
  - Group: $AC = No$ (1 row: [15])
    - Go Out = Yes: 0
    - Go Out = No: 1 (row 15)
- $H(S_{F=No,\,AB=Mid,\,AC=Yes}) = -\frac{2}{3}log_2\frac{2}{3} - \frac{1}{3}log_2\frac{1}{3} = 0.9183$
- $H(S_{F=No,\,AB=Mid,\,AC=No}) = -\frac{0}{1}log_2\frac{0}{1} - \frac{1}{1}log_2\frac{1}{1} = 0$
- $IG(S_{F=No,\,AB=Mid,}, AC) = 1 - \frac{3}{4}(0.9183) - \frac{1}{4}(0) = 0.3113$

Feature C

- Counts:
  - Group: $C = London$ (1 row: [5])
    - Go Out = Yes: 1 (row 5)
    - Go Out = No: 0
  - Group: $C = Birmingham$ (2 rows: [8, 15])
    - Go Out = Yes: 1 (row 8)
    - Go Out = No: 1 (row 15)
  - Group: $C = Oxford$ (1 row: [22])
    - Go Out = Yes: 0
    - Go Out = No: 1 (row 22)
- $H(S_{F=No,\,AB=Mid,\,C=London}) = -\frac{1}{1}log_2\frac{1}{1} - \frac{0}{1}log_2\frac{0}{1} = 0$
- $H(S_{F=No,\,AB=Mid,\,C=Birmingham}) = -\frac{1}{2}log_2\frac{1}{2} - \frac{1}{2}log_2\frac{1}{2} = 1$
- $H(S_{F=No,\,AB=Mid,\,C=Oxford}) = -\frac{1}{1}log_2\frac{1}{1} - \frac{0}{1}log_2\frac{0}{1} = 0$
- $IG(S_{F=No,\,AB=Mid}, C) = 1 - \frac{1}{4}(0) - \frac{2}{4}(1) - \frac{1}{4}(0) = 0.5$

Splitting on Feature C next, as IG = 0.5 is highest.

Level 3: Split on C (From F=No, AB=Mid)

Branch 1b.2b.3a: C=London (1 instance)

Indices in this branch: [5]

Class Counts

- Go Out = Yes: 1 (row 5)
- Go Out = No: 0

LEAF NODE!

Branch 1b.2b.3b: C=Birmingham (2 instances)

Indices in this branch: [8, 15]

Class Counts

- Go Out = Yes: 1 (row 8)
- Go Out = No: 1 (row 15)

Entropy

$$H(S_{F=No,\,AB=Mid}) = -[\tfrac{1}{2}log_2\tfrac{1}{2} + \tfrac{1}{2}\,log_2\tfrac{1}{2}] = 1$$

Feature AC

- Counts:
  - Group: $AC = Yes$ (1 row: [8])
    - Go Out = Yes: 1 (row 8)
    - Go Out = No: 0
  - Group: $AC = No$ (1 row: [15])
    - Go Out = Yes: 0
    - Go Out = No: 1 (row 15)
- $H(S_{F=No,\,AB=Mid,\,C=Birmingham,\,AC=Yes}) = -\frac{1}{1}log_2\frac{1}{1} - \frac{0}{1}log_2\frac{0}{1} = 0$
- $H(S_{F=No,\,AB=Mid,\,C=Birmingham,\,AC=No}) = -\frac{0}{1}log_2\frac{0}{1} - \frac{1}{1}log_2\frac{1}{1} = 0$
- $IG(S_{F=No,\,AB=Mid,\,C=Birmingham}, AC) = 1 - \frac{1}{2}(0) - \frac{1}{2}(0) = 1$

Splitting on Feature AC next, as it is our only option

Level 4: Split on AC (From F=No, AB=Mid, C=Birmingham)

Branch 1b.2b.3b.4a: AC=Yes (1 instance)

Indices in this branch: [8]

Class Counts

- Go Out = Yes: 1 (row 8)
- Go Out = No: 0

LEAF NODE!

Branch 1b.2b.3b.4a: AC=No (1 instance)

Indices in this branch: [15]

Class Counts

- Go Out = Yes: 0
- Go Out = No: 1 (row 15)

LEAF NODE!

Branch 1b.2b.3c: C=Oxford (1 instance)

Indices in this branch: [22]

Class Counts

- Go Out = Yes: 0
- Go Out = No: 1 (row 22)

LEAF NODE!

Branch 1b.2c: AB=Low (3 instances)

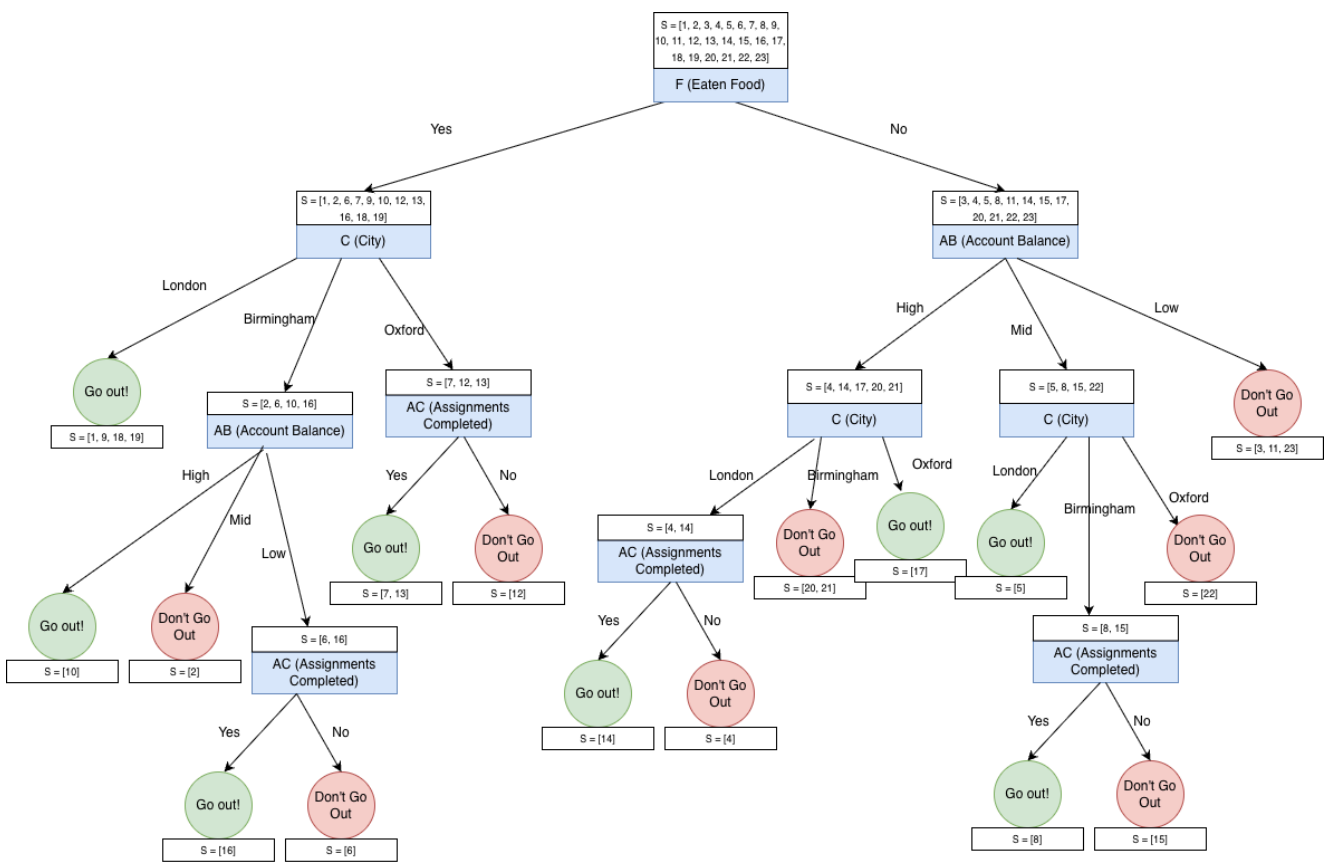
Indices in this branch: [3, 11, 23]

Class Counts

- Go Out = Yes: 0
- Go Out = No: 3 (rows 3, 11, 23)

LEAF NODE!

Part D: Coloured Decision Tree Diagram



I employ rectangles to represent decision nodes (feature tests), circles to represent lead nodes. Blue is used to color the decision nodes, green is used to color “Go Out = Yes” leaves, and red is used for “Go Out = No” leaves.

Part D: Narrative Explanation and Classification of Test Examples

[285 words]

Narrative:

The decision tree predicts whether I will go out for a pint. The root node tests Eaten Food (F):

If **food has been eaten**, the tree next considers City (C). In London, the model directly predicts going out. In Birmingham, the decision depends on Account Balance (AB), where if we have a high balance, we go out, a mid balance results in us staying home, and a low balance considers Assignments Completed (AC), where we go out if we’ve done the assignments. A similar check is done in Oxford.

If **food has not been eaten**, the tree instead splits on Account Balance (AB). A low balance directly predicts not going out. For mid or high balances, the tree further examines **City**, and in some cases Assignments Completed, before making a final prediction.

New Test Instances:

(1) AC = No, AB = High, C = Oxford, F = No

- Root: *F = No* → “No” branch
- AB: *High* → “High” branch
- City: *Oxford* → “Oxford” branch
- Prediction: *Go Out = Yes*
- *Limitation*: This prediction arises because the corresponding leaf node was trained on only 1 sample (where AC = Yes), causing the model to prioritise account balance over unfinished assignments.

(1) AC = No, AB = Low, C = London, F = Yes

- Root: *F = Yes* → “Yes” branch
- City: *London* → “London” branch
- Prediction: *Go Out = Yes*
- *Limitation*: This leaf node only uses examples where either AC=yes, or AB = mid or high. It fails to consider the case where we have not completed assignments, and we have a low balance, and recommends still going out.

Key Limitation Cause: The tree is trained on a small dataset, leading to overfitting and leaf nodes based on very few examples (often 1). This can produce unintuitive predictions.

Question 3: Neural Network Mathematics & Dynamics

Part 1:

Weights are being initialized with $w \sim N(0, 1)$. Since weights are centered at 0, approximately 50% of the activation inputs in the network will be negative.

$ReLU(x) = max(0, x)$), so the output for each negative input will be 0. For each node in each layer, ReLU outputs 0 for 50% of inputs, halving the probability mass.

The likelihood of a signal surviving 10 consecutive layers of 50% "pruning" is mathematically miniscule ($0.5^{10} \approx 0.00098$), which is why in this scenario, we see all the activation outputs are exactly 0.0.

Part 2:

Our target outputs are all negative ($y < 0$). Our final ReLU outputs must be non-negative ($\hat{y} \geq 0$).

- Loss Function (MSE): $L = \frac{1}{2}(\hat{y} - y)^2$
- Gradient with Respect to weights:

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial w}$$

- Gradient with respect to $\hat{y}$: $\frac{\partial L}{\partial \hat{y}} = (\hat{y} - y)$

  Note: Since $y < 0$ and $\hat{y} \geq 0$, this is always positive and non-zero

- Activation Gradient. For ReLU: $\frac{\partial \hat{y}}{\partial z} = \begin{cases} 1 & \text{if } z > 0 \\ 0 & \text{if } z \leq 0 \end{cases}$

- Calculation: Gradient with respect to weights
  - As soon as a negative value for input z is predicted, ReLu(z) becomes 0.
  - So,

$$\frac{\partial L}{\partial w} = (\hat{y} - y) \cdot 0 \cdot \frac{\partial z}{\partial w} = \mathbf{0}$$

Final answer: the gradient of the loss with respect to the network's weights is equal to **0**.

Part 3:

I would have stopped training the model at epoch 25, where the validation loss reaches its minimum, after which it begins to increase. This increase is indicative of the model overfitting to the training data, and we want to stop before that point.

Part 4:

The most likely cause is a very small validation dataset. The loss stays flat until the model's learning finally "tips" a specific validation example into a better-predicted state, causing a sudden, visible drop in the validation loss.

Part 5:

Without any non-linear activation functions like ReLU between the layers, a 10-layer linear MLP is equivalent to a single affine transformation.

Proof:
Each dense layer i computes:

$$h_i = W_i h_{i-1} + b_i, \text{ where } h_0 = x$$

So for example:

- Layer 1: $h_1 = W_1 x + b_1$
- Layer 2: $h_2 = W_2 h_1 + b_2$
- Layer 3: $h_3 = W_3 h_2 + b_3$

h_1 can be substituted into the equation for h_2:

- $h_2 = W_2(W_1 x + b_1) + b_2 = (W_2 W_1)x + (W_2 b_1 + b_2)$

We can generalize this substitution up until the 10th layer:

- $(W_{10} W_9 \ldots W_1)x + bias\ terms$

Since the product of any amount of linear transformation matrices is also a linear transformation matrix, we can effectively say:

- $W_{effective} = W_{10} W_9 \ldots W_1$
- $b_{effective} = combined\ bias\ terms$

So the output of the 10-layer network simplifies to:

$$W_{effective} x + b_{effective},$$

which satisfies the conditions of an affine transformation.

Question 4: Planning Logic

Part A: Action Preconditions and Effects in First-order Logic

1) Take whole Apple
   - Preconditions: $Available(Apple)$
   - Effect: $In_Hand(Apple) \ \wedge \ \neg Available(Apple)$
2) Peel Apple
   - Preconditions: $Is_Clean(Apple) \ \wedge \ \neg Is_Peeled(Apple) \wedge In_Hand(Apple) \ \wedge In_Hand(Knife)$
   - Effect: $Is_Peeled(Apple)$
3) Cut Apple
   - Preconditions: $Is_Peeled(Apple) \ \wedge \ \neg Is_Cut(Apple) \wedge In_Hand(Apple) \ \wedge In_Hand(Knife)$
   - Effect: $Is_Cut(apple)$
4) Put down Knife
   - Preconditions: $In_Hand(Knife)$
   - Effect: $Available(Knife) \ \wedge \ \neg In_Hand(Knife)$
5) Add chopped apple to the container
   - Preconditions: $In_Hand(Apple) \ \wedge Empty(Container)$
   - Effect: $Contains(Container, Apple) \ \wedge \ \neg Empty(Container) \ \wedge \ \neg In_Hand(Apple)$

Part B: FOL to CNF

1) <u>The action of peeling an apple.</u>

Forming Implication:

$Is_Clean(Apple) \ \wedge \ \neg Is_Peeled(Apple) \wedge In_Hand(Apple) \ \wedge In_Hand(Knife) \ \Rightarrow Is_Peeled(Apple)$

Eliminate Implication: A → B ⇒ ¬A ∨ B

$\neg(Is_Clean(Apple) \ \wedge \ \neg Is_Peeled(Apple) \wedge In_Hand(Apple) \ \wedge In_Hand(Knife)) \ \vee \ Is_Peeled(Apple)$

DeMorgan's Laws: ¬(A ∧ B) ⇒ ¬A ∨ ¬B

$\neg Is_Clean(Apple) \ \vee Is_Peeled(Apple) \ \vee \ \neg In_Hand(Apple) \vee \ \neg In_Hand(Knife) \ \vee Is_Peeled(Apple)$

Remove Repeated Clauses: Is_Peeled(Apple)

CNF: $\neg Is_Clean(Apple) \ \vee Is_Peeled(Apple) \ \vee \ \neg In_Hand(Apple) \vee \ \neg In_Hand(Knife)$

2) <u>The action of cutting an apple.</u>

Forming Implication:

$Is_Peeled(Apple) \ \wedge \ \neg Is_Cut(Apple) \wedge In_Hand(Apple) \ \wedge In_Hand(Knife) \ \Rightarrow Is_Cut(Apple)$

Eliminate Implication: A → B ⇒ ¬A ∨ B

$\neg(Is_Peeled(Apple) \ \wedge \ \neg Is_Cut(Apple) \wedge In_Hand(Apple) \ \wedge In_Hand(Knife)) \ \vee \ Is_Cut(Apple)$

DeMorgan's Laws: ¬(A ∧ B) ⇒ ¬A ∨ ¬B
$\neg Is_Peeled(Apple) \vee Is_Cut(Apple) \vee \neg In_Hand(Apple) \vee \neg In_Hand(Knife) \vee Is_Cut(Apple)$

Remove Repeated Clauses: Is_Cut(Apple)
CNF: $\neg Is_Peeled(Apple) \vee Is_Cut(Apple) \vee \neg In_Hand(Apple) \vee \neg In_Hand(Knife)$

Question 5: CIFAR-100 Semantic Expansion

[498 words]

Methodology

Developing this embedding consisted of two phases:

1. Improving existing embedding
2. Inserting CIFAR-100 words into embedding

In each phase, the work consisted of:

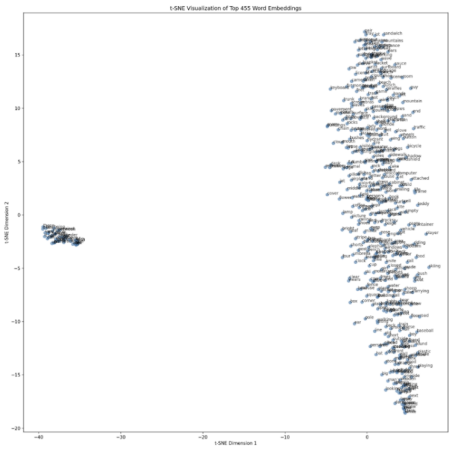
a) Key Logic Improvements expected to be unambiguously beneficial
b) Grid searches across different parameters

The "best" settings from each grid search are carried forward into subsequent phases and grid searches, resulting in our final embedding space.

Evaluation

Two metrics were formulated to evaluate embeddings.

- **Signal-to-Noise Ratio:** The mean intra–CIFAR superclass similarity minus the mean similarity between 100 non-stopwords. Maximizing this discourages embedding collapse and encourages intra-superclass similarity.
    - Restricting the mean to non-stopwords avoids favouring solutions where stopwords are far from content words while all non-stopwords collapse together:



- **CIFAR-Neighbours:** The average number of nearest neighbours that are in the same CIFAR superclass. Higher values indicate better semantic clustering of related concepts.

Phase 1: Improving existing embedding space

A) Key Logic Improvements

A) Excluding All Contexts from Negatives: Prevents contradictory training signals.

B) Only include stopwords as context if target is a stopword: Prevents commonly used words from being too close to non-stopwords, also encourages a separate stopword cluster.

B) Grid Searches: Hyperparameter Tuning

Grid Search 1: Architecture Parameters

Search Space:

Parameter	Values Tested
Embedding Dimension	[128, 256]
Number of Negatives	[8, 13, 18, 23]
Context Size	[1, 2, 3]

Fixed Parameters:

```
fixed_params = {
    'batch_size': 1024,
    'epochs': 20,
    'learning_rate': 0.001,
    'validation_fraction': 0.1,
    'dropout': 0.2,
    'weight_decay': 1e-4,
    'label_smoothing': 0.1,
    'exclude_all_contexts': True,
    'patience': 5,
    'device': None
}
```

Results

Rank	Dimension	Negatives	Context Size	Metric Value
Top 6 Embeddings by Signal-to-Noise Ratio:				
1	256	8	1	0.0243
2	128	13	1	0.0192
3	256	18	1	0.0188
4	256	13	1	0.0112
5	128	8	1	0.0095
6	128	23	1	0.0049
Top 6 Embeddings by CIFAR-Neighbours Metric:				
1	256	18	1	0.3077
2	128	23	1	0.2308
2	256	13	1	0.2308
2	256	8	1	0.2308
2	256	8	2	0.2308
3	128	13	1	0.1923

Top Embeddings:

- Signal-to-Noise: [Dimension: 256, Negatives: 8, Context: 1]
- CIFAR-Neighbours: [Dimension: 256, Negatives: 18, Context: 1].

I prioritized CIFAR-Neighbours performance, as that is more indicative of semantic similarity.

The architecture settings we will carry forward: [Dimension: 256, Negatives: 18, Context: 1]

Grid Search 2: Regularization Parameters

Search Space:

Parameter	Values Tested
Learning Rate	[0.01, 0.03, 0.05]
Dropout	[0.1, 0.2, 0.3]
Weight Decay	[1e-5, 1e-4, 1e-3]
Label Smoothing	[0.0, 0.1, 0.2]

Fixed Parameters:

```
fixed_params = {
    'embedding_dim': 256,
    'num_negative': 18,
    'context_size': 1,
    'batch_size': 1024,
    'epochs': 20,
    'validation_fraction': 0.1,
    'exclude_all_contexts': True,
    'patience': 5,
    'device': None
}
```

Results

Rank	Learning Rate	Dropout	Weight Decay	Metric Value
Top 6 Embeddings by Signal-to-Noise Ratio:				
1	0.03	0.2	1.00E-04	0.3053
2	0.03	0.1	1.00E-05	0.2913
3	0.03	0.2	1.00E-05	0.2889
4	0.03	0.1	1.00E-03	0.2856
5	0.03	0.1	1.00E-04	0.2731
6	0.03	0.1	1.00E-05	0.2665
Top 6 Embeddings by CIFAR-Neighbours Metric:				
1	0.03	0.1	1.00E-05	1.2692
2	0.05	0.2	1.00E-04	1.0769
3	0.03	0.2	1.00E-04	1
4	0.03	0.2	1.00E-04	0.9615
5	0.05	0.3	1.00E-04	0.9231
6	0.05	0.3	1.00E-03	0.8846

Two embeddings appear in the top 6 for both metrics:

- [Learning Rate: 0.3, Dropout: 0.2, Weight Decay: 1e-4. Label Smoothing: 0]
- [Learning Rate: 0.3, Dropout: 0.1, Weight Decay: 1e-5. Label Smoothing: 0.1].

I select the first embedding's settings due to a higher mean rank between the two metrics.

The final settings for the pre-insertion embedding are: Dimension: 256, Negatives: 18, Context: 1 (Phase 1), and Learning Rate: 0.3, Dropout: 0.2, Weight Decay: 1e-4. Label Smoothing: 0 (Phase 2)

Phase 2: Inserting CIFAR-100 into Embedding Space

A) Key Logic Improvements

1. **Creating anchors for new words:** Used in fitness function
2. **Building Contexts for words for fitness function:**
    a. Handle CIFAR-100 words that do not appear in vg_text due to formatting or rarity
        i. Solution: Create a copy of the vg_text file, and replace similar words with their CIFAR counterpart (e.g. aquarium -> aquarium_fish, rat -> shrew).
    a) Only include stopwords as context if target is a stopword
    b) Limit context searching to sentence boundaries

B) Grid Searches: Fitness Function Modifications

Grid Search 1: Anchor Experiments

Tested using many anchors per word (4-5) versus a limited set (1-2), to compare their impact on performance.

Results

Anchors	Signal-to-Noise	CIFAR-Neighbours
Limited	0.2743	1.5698
Many	0.1703	1.4767

The embedding using limited anchors performed better in both metrics, so we will carry it forward.

Grid Search 2: Fitness Function Weights

Different combinations of fitness function weights were tested:

Search Space:

Corpus Similarity	**Magnitude Matching**	**Anchor Similarity**
0.2	0.2	0.6
0.6	0.15	0.25
0.5	0.15	0.35
0.5	0.25	0.25
0.4	0.25	0.35
0.3	0.3	0.4

Results:

Corpus Similarity	Magnitude Matching	Anchor Similarity	Signal-to-Noise	CIFAR-Neighbours
0.2	0.2	0.6	0.5019	1.6163
0.3	0.3	0.4	0.4286	1.6047
0.4	0.25	0.35	0.3264	1.3837
0.5	0.15	0.35	0.2997	1.5465
0.6	0.15	0.25	0.2728	1.5698
0.5	0.25	0.25	0.2683	1.5116

The embedding with [Corpus: 0.2, Norm: 0.2, Anchor: 0.6] had the highest rank for both metrics, so this will be our final embedding.